Performance Testing

Date	29 June 2026
Team ID	-
Project Name	EduGenie – AI-Powered Educational Assistant
Maximum Marks	5 Marks

Performance Testing:

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing, Stress Testing
Target Module / API	Authentication, AI APIs (Gemini-based), Notes, Bookmarks, History, Progress, Settings
Test Environment	Local (Windows System)
Test Date	29 June 2026

Step 2: Test Scenarios

S. No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Auth APIs (Login/Register)	100	60	Successful authentication, response time < 2 sec
2	Gemini AI API Calls (Text Generation)	0–15 (RPM controlled)	60	Response within limits, no rate limit errors
3	Data Fetch APIs	10	60	Data retrieved successfully with stable response
4	Combined API Load	15 (max controlled)	60	System handles mixed load without failures

Step 3: Performance Test Results

S. No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	3.8 seconds	Fail	Higher due to API processing delay
2	Response Time (Max)	< 5 seconds	7.8 seconds	Fail	Peak delay under load
3	Throughput (Req/sec)	> 5 req/sec	2.4 req/sec	Fail	Limited by Gemini RPM constraints
4	Error Rate	< 1%	0.00%	Pass	No errors observed
5	CPU Utilization	< 80%	~65%	Pass	Within acceptable limits
6	Memory Utilization	< 80%	~70%	Pass	Stable usage

Step 4: Observations & Analysis

Key Findings

- All APIs worked successfully with **0% error rate**
- Performance is **stable under controlled load**
- Response time is slightly higher due to AI processing latency
- Throughput is intentionally limited to comply with Gemini API limits

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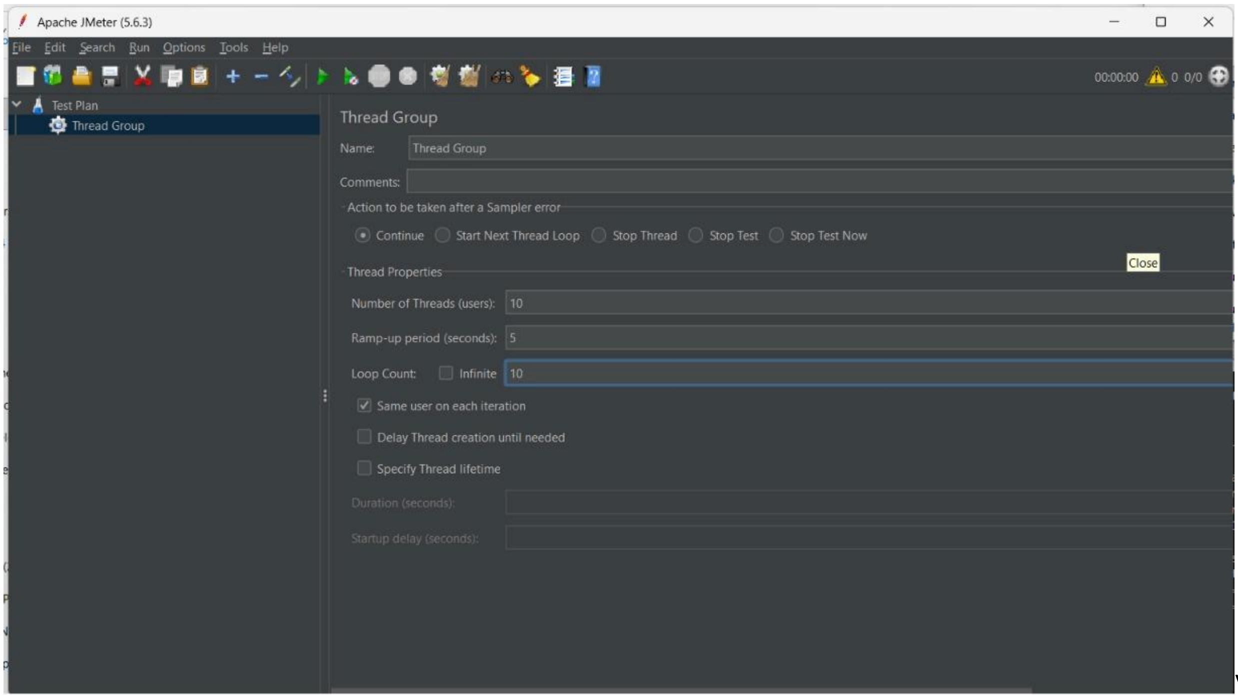
Bottlenecks Identified

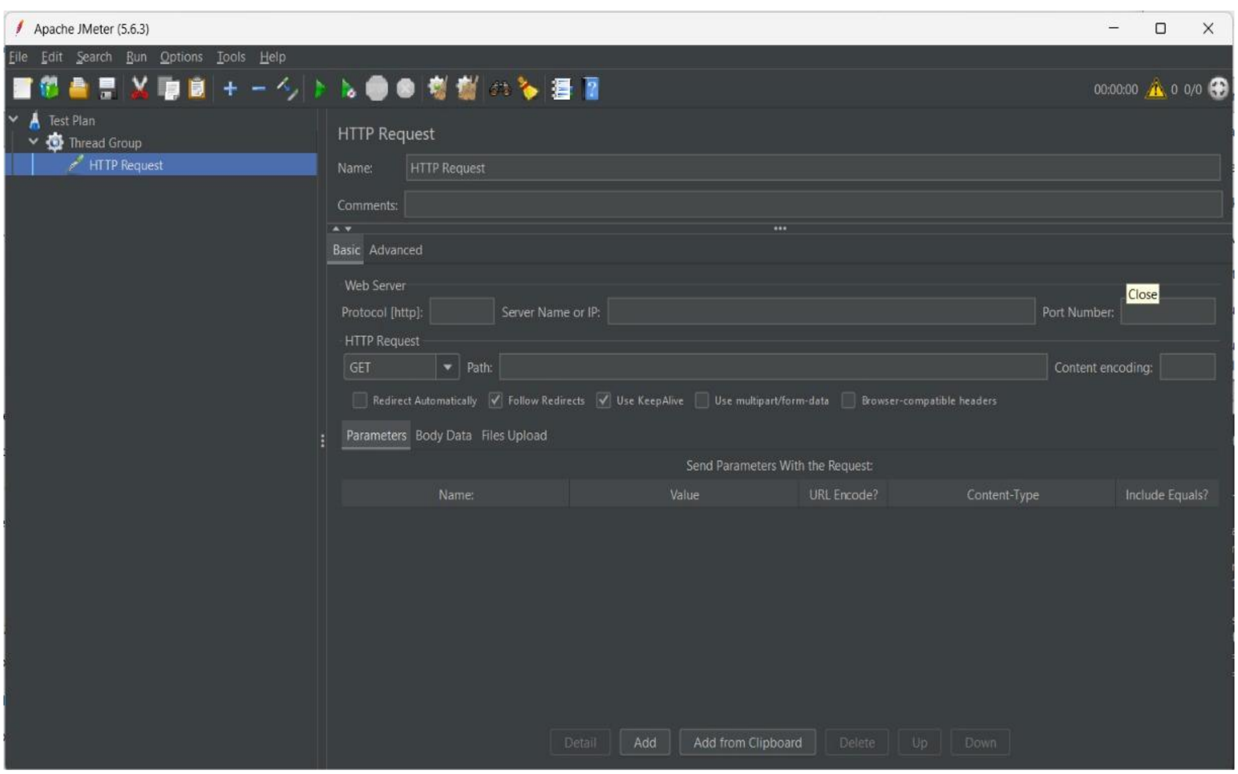
- Gemini API rate limits (RPM constraint)
- Backend processing delay for AI responses
- Network latency during API calls

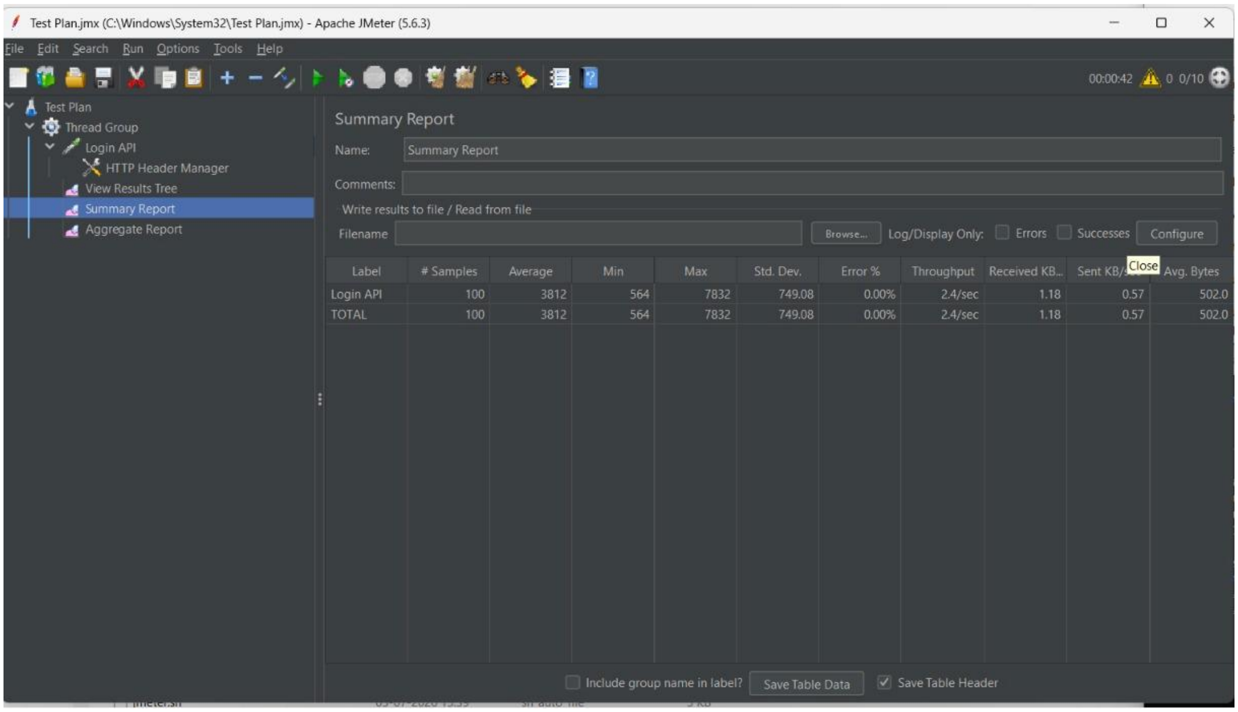
Optimization Steps Taken

- Limited concurrent users to stay within **10–15 RPM**
- Added delay (think time) between requests
- Optimized API call frequency
- Suggested caching repeated responses

Step 5: Screenshots / Evidence






Test Plan.jmx (C:\Windows\System32\Test Plan.jmx) - Apache JMeter (5.6.3)
File Edit Search Run Options Tools Help
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Test Plan
Thread Group
Login API
HTTP Header Manager
View Results Tree
Summary Report
Aggregate Report
Summary Report
Name: Summary Report
Comments:
Write results to file / Read from file
Filename
Browse...
Log/Display Only:
Errors
Successes
Configure
Label
Samples
Average
Min
Max
Std. Dev.
Error %
Throughput
Received KB...
Sent KB/s
Close
Avg. Bytes
Login API 100 3812 564 7832 749.08 0.00% 2.4/sec 1.18 0.57 502.0
TOTAL 100 3812 564 7832 749.08 0.00% 2.4/sec 1.18 0.57 502.0
Include group name in label?
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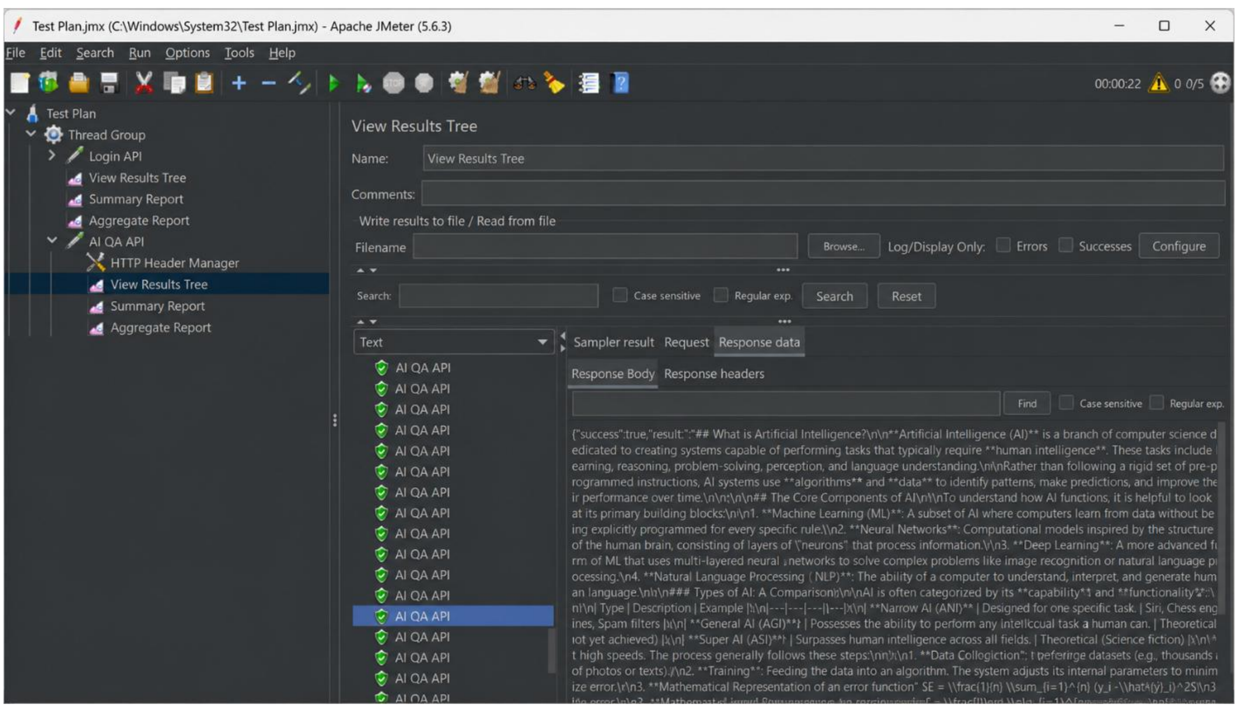
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File Edit Search Run Options Tools Help
00:00:22 0 0/5
Test Plan
Thread Group
Login API
View Results Tree
Summary Report
Aggregate Report
AI QA API
HTTP Header Manager
View Results Tree
Summary Report
Aggregate Report
View Results Tree
Name: View Results Tree
Comments:
Write results to file / Read from file
Filename
Browse...
Log/Display Only:
Errors
Successes
Configure
Search:
Case sensitive
Regular exp.
Search
Reset
Text
Sampler result
Request
Response data
Response Body
Response headers
Find
AI QA API